

STARTUP ZONE

Hummingbird: Hawk-Eye Vision For Surveillance Drones



Hummingbird is a series of advanced image-driven products for aerial platforms

Superior optical systems are mandatory for defence devices like weapons and drones. Optoelectronics for weapons is a complex domain that has not been explored by many. India has been highly-dependent on Russia and other countries for most optoelectronic components.

Costs are quite high if a nation is dependent on others to take care of its own defence. Looking at this glaring lack of ecosystem in this field in India, Tonbo Imaging India stepped up to deliver.

Ankit Kumar, co-founder and chief technology officer, Tonbo Imaging India, says, “When we were building Tonbo in 2008, most companies were heavily focussed on e-commerce, software or outsourcing services. The idea was to create electro-optic systems for India as well as other emerging markets and, in the process, kickstart an ecosystem around it here. Our founding members have a good understanding of hardware, software and the sciences around any engineering field—mechanical, electrical,

electronics, optics and so on—to work together and build these systems.”

One of Tonbo’s most interesting products is the Hummingbird series, which is a family of advanced gyro-stabilised multi-band imaging gimbal systems for unmanned aerial vehicles (UAVs).

How Hummingbird works

In simple terms, Hummingbird is a series of advanced image-driven products. These are cameras installed inside a ball turret package, and can be installed in drones or similar aerial platforms to lock on to a target. Hummingbird maintains line-of-sight on a target when it flies. The imaging system is gyro-stabilised, which means, irrespective of the motion of the aerial platform, it maintains focus on the target till it catches

up with it. The system has a layer of advanced technologies, sensors and complex program algorithms. Using these, through real-time video image processing, it can lock on to and track the target automatically.

Kumar explains, “Closed-loop algorithms run on the gimbal to maintain a lock on the target. Control systems inside the gimbal contain inertial sensors that sense the motion of the platform, and produce counter-motion accordingly to keep the video stable, irrespective of the aerial motion. The system is a complete integration of camera sensors, inertial sensors, motors and image processing capabilities.” The imaging system includes infrared (IR) and thermal imagers that are useful in low-light conditions.

Regarding the tools used for developing the system, Kumar says, “We are using Qualcomm Snapdragon as the platform to create our video-image pipeline. The backend is native to hardware, utilising ARM-based coding.” The products contain about 70 per cent Made in India components. Hummingbird can be used in small or lightweight drones, which can be deployed in urban or border areas to



Tracking and geo-referencing target by Hummingbird on drones